SUBJECT	Lower Secondary Maths Year 8
TOPIC	Module Three
ASSIGNMENT	Three
STUDENT NAME	

FEEDBACK AND REFLECTION:

- ☐ I have read the feedback on my last assignment carefully.
- ☐ I have considered whether any of that feedback is relevant to this assignment.

One thing I will try to do either the same or differently in this assignment as a result of my previous feedback is: ____

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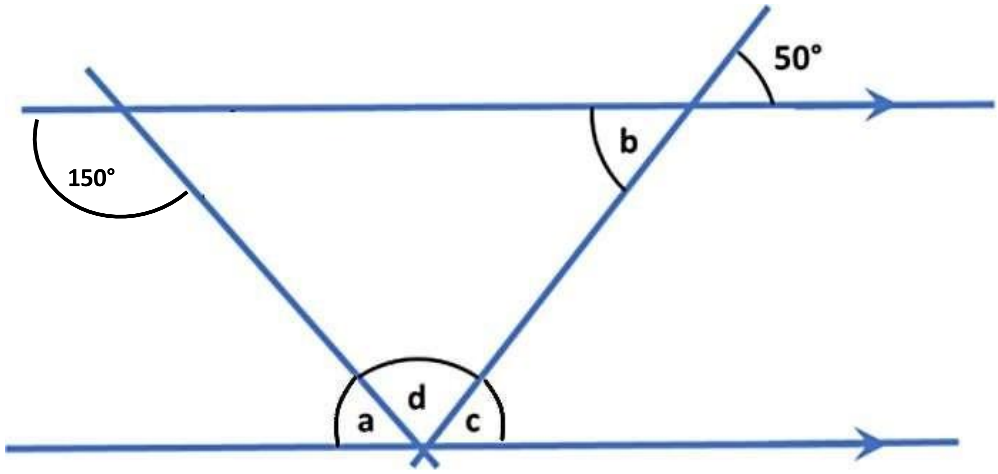
ASSIGNMENT GUIDANCE

- Make sure you write your full name clearly in the box at the top of this page.
- Please **write** your answers **by hand** in **black ink** on this question paper unless you have obtained specific permission to submit typed work.
- This is not a timed assignment. We estimate that this assignment may take you between 30 and 40 minutes. However, if you need to take longer, then please do so.
- You are not expected to complete your assignments without referring to your notes or coursebook.
- In order for maximum marks to be awarded, for any question worth more than one mark, please show your workings to make your method clear.
- Calculators are **not** to be used in this assignment.

LS Maths Assignment Three

1. Find the values of the unknown angles a, b, c and d in the diagram below, which has not been drawn to scale. Show clearly how you found each angle, using the correct mathematical terms (e.g. alternate angles) in your explanations.



a =　　　　b =　　　　c =　　　　d =

a.

b.

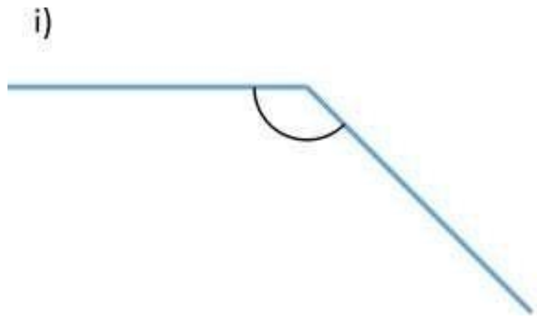
c.

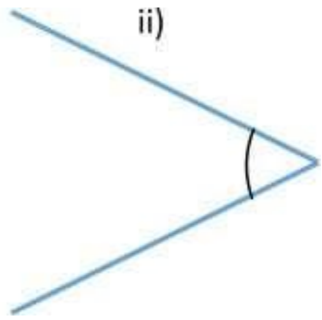
d.

(8 marks)

2.

a. Measure these angles using a protractor.





Angle i) ______________ Angle ii) _____________ (2 marks)

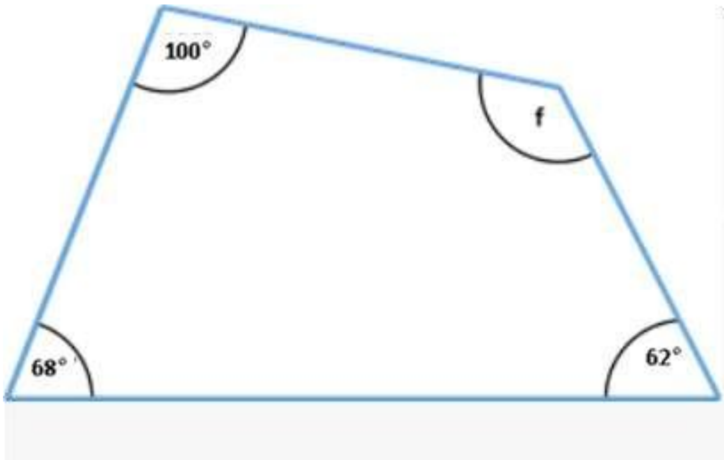
3. A regular polygon has an interior angle of 172°. Find the exterior angle and work out how many sides it has, making your method clear.

Exterior angle = Number of sides =

(4 marks)

4.

a. Show how you could calculate the angle marked by the letter *f*: (1 mark)



b. Carefully measure each side and write the lengths on the diagram. Use this information to construct the above quadrilateral as accurately as you can using a ruler and a protractor. (You should construct this afresh without copying/tracing over the shape above.)

(4 marks)

(Total 5 marks)

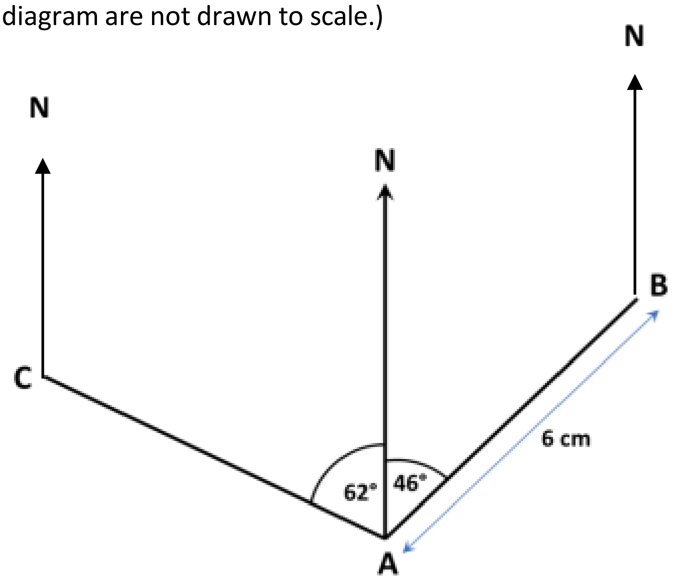
5. A map has a scale of 1:250 000. How many centimetres on the map is a journey of 40 km?

   Remember to show your workings.

(3 marks)

6. Look at the diagram below, which shows the position of three towns, A (Alton), B (Bolton) and C (Carlton). The scale used in the diagram is 1 cm to 2.5 km.

(The angles in the diagram are not drawn to scale.)



a. What is the real-life distance from Alton to Bolton? Remember to make your method clear. (2 marks)

b. What is the bearing of Alton from Bolton?

i. Mark the angle to be found on the diagram above, labelling it b. (1 mark)

ii. Find the value of this bearing (1 mark)

c. What is the bearing of Alton from Carlton?

i. Mark the angle to be found on the diagram above, labelling it c. (1 mark)

iii. Find the value of this bearing (1 mark)

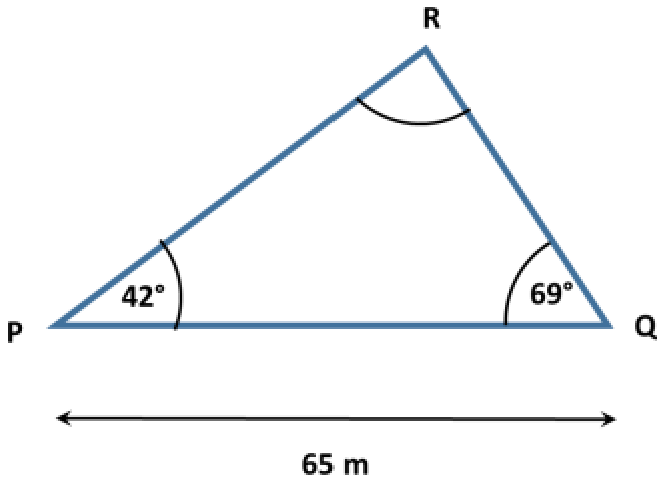
(Total 6 marks)

7. A boat starts at a point A. It travels a distance of 7 km on a bearing of 135° to point B. From B it travels 12 km on a bearing of 250° to point C. Using a scale of 1 cm: 1 km, draw a diagram to show these bearings and journeys.

(4 marks)

8.

a. Construct the following using a ruler and protractor, using the measurements as indicated on the triangle. (**Note**: we have intentionally not drawn our triangle to scale!)
Use a scale of 1cm to represent 10m.



(2 marks)

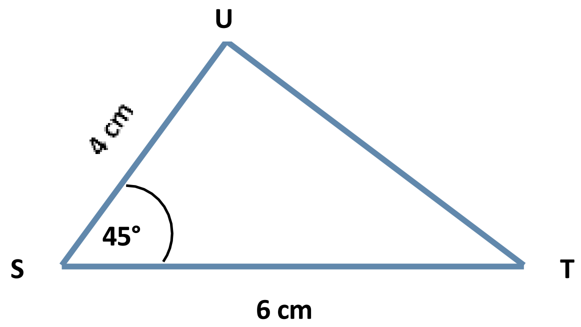
b. Measure angle R using a protractor. State its value. (1 mark)

c. What is the specific name given to this type of triangle? (1 mark)

(Total 4 marks)

9.

a. Construct the following using a ruler and protractor, using the measurements as indicated on the triangle. (**Note**: we have intentionally not drawn our triangle to scale!)
Use a scale of 1cm to represent 10m.



(2 marks)

b. Using a protractor, find the value of angle U. (1 mark)

c. Explain how we could **calculate** the unknown angle T. (1 mark)

(Total 4 marks)

TOTAL FOR ASSIGNMENT 40 MARKS